Selection on Fertility, and the Environmental Decline That Would Cancel It

A compositional mechanism inside a validated 237-country projection to 2150

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Version 1.0.0 • 15 August 2026

Abstract

Long-run population projections model the national fertility rate as a time series that reverts toward a long-run level. That representation has no composition in it. But completed family size varies widely within every population, and it is transmitted imperfectly from parents to children, so the next generation is drawn disproportionately from the larger families. This is selection; its direction on fertility is upward; and it is absent from the standard projection machinery by construction.

We add a composition axis to a single-year, two-sex cohort-component projection covering 237 countries and validated against the United Nations' own zero-migration variant to 0.001% of world population at 2100. With every fertility propensity set to one, the extended engine reproduces the ordinary one to machine precision, so any difference in what follows is the mechanism rather than a second implementation of the arithmetic. The mechanism's two mainstream parameters are taken from evidence that is not the series they are meant to explain: the dispersion of completed family size across 19 low-fertility countries (median coefficient of variation 0.570) and the parent–child correlation in completed family size reported in the literature (0.15). Their product with the squared dispersion is the model's one-generation response, and it equals an observed covariance rather than an assumed heritability.

Selection is real but modest at the century scale. It raises world population in 2150 from 8.54 to 10.36 billion and lifts observed fertility 17.4% above the environment path that produced it. Named high-fertility groups, which dominate popular discussion of this mechanism, add a further 0.05 billion: the unlabelled variation inside ordinary populations does almost all of the work.

Because selection and the fertility environment both act multiplicatively, they are directly commensurable, and our primary result is a boundary rather than a preferred number. An additional uniform decline in the fertility environment of 1.52% per decade after 2050 exactly cancels measured mainstream selection by 2150 (90% range 1.51% to 1.53% across 50 paired stochastic migration paths); across the empirical range of the two measured parameters the threshold runs from 0.33% to 4.49% per decade. The unmeasurable quantity in this problem is thereby converted from a free parameter that sets the headline into a threshold that can be argued about.

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1 Introduction

At a 150-year horizon, almost nobody alive at the start is alive at the end. A projection to 2150 from a 2024 base year is therefore not really a statement about today's population. It is a statement about the process that generates fertility, iterated four or five times, and almost everything else washes out. Freezing today's fertility rates and letting the arithmetic run produces about 53 billion people; the United Nations' medium assumptions produce about 8.78 billion. The base data are the same in both cases. The difference is entirely a claim about what fertility does next.

Conventional long-run projections make that claim in a specific form. Total fertility is treated as a national time series with a random walk down through the demographic transition and a probabilistic reversion toward a long-run level afterwards [Alkema et al., 2011, Raftery et al., 2012, Ševčíková et al., 2016]. The United Nations [United Nations Population Division, 2024a] and the Institute for Health Metrics and Evaluation [Vollset et al., 2020] differ in their estimated levels, but they agree on the shape of the object being modelled: one number per country per year, evolving by its own history.

That object has no composition in it, and composition is not a detail here. In every population that has been measured, women differ a great deal in how many children they end up having. Across 19 low-fertility countries with completed cohort data, the coefficient of variation of completed family size has a median of 0.570, meaning the standard deviation is more than half the mean. And that variation is transmitted: adult children of larger families have somewhat larger families themselves, with reported parent–child correlations usually between 0.1 and 0.2 [Murphy, 1999, 2013].

Put those two facts together and something follows that a national-rate model cannot express. The parents of the next generation are not a random sample of this generation — they are sampled in proportion to how many children they have. Whatever disposition made a woman have four children rather than one is therefore over-represented among the people who raise the next cohort, and if that disposition is partly transmitted, the composition of the population shifts toward it. This is selection in the plain sense used since Fisher [1930], and its direction on fertility is always upward, because fertility is simultaneously the trait and the fitness measure. A model that projects only the national mean assumes this force away rather than estimating it to be small.

What is new here

That selection acts on fertility is not a new observation. Kolk et al. [2014] show that intergenerational fertility correlations can prevent low fertility from persisting; Burger and DeLong [2016] argue that treating the fertility decline as permanent is unjustifiable on evolutionary grounds; and Collins and Page [2019] apply the breeder's equation to national fertility and conclude that global population stabilisation this century is unlikely. There is also direct evidence that contemporary human populations are under measurable selection on fertility-linked traits [Byars et al., 2010, Beauchamp, 2016, Kong et al., 2017].

The sharpest objection to that literature is also the most useful. Arenberg et al. [2022] point out that a model built only from heritability has no absolute fertility level in it: it can describe who becomes more common without being able to say whether the population grows, and the empirical fact that higher-fertility subgroups are themselves trending toward replacement has nowhere to enter. Their condition is clean — a high-fertility type shrinks whenever its retention times its fertility falls below one — and it cannot be evaluated in a model that carries no fertility levels.

This paper's contribution is to make that evaluation possible, and then to make the result useful despite the parameter nobody can measure. Specifically:

1. **The mechanism runs inside a real projection.** Selection is added as an extra axis on a single-year, two-sex cohort-component engine for 237 countries carrying the UN's own age-specific fertility, survival and migration schedules (Section 4). Absolute levels are present throughout, so the Arenberg et al. [2022] condition is enforced by the arithmetic rather than assumed away. With every propensity set to one the extended engine reproduces the ordinary one to machine precision, which is what makes any difference in the results attributable to the mechanism.
2. **The mechanism's parameters are not fitted to the series they explain.** The two mainstream parameters are the dispersion of completed family size and the parent–child correlation in it, both taken from independent sources (Section 3). We show that their combination reproduces an *observed covariance*, which is a much weaker and more defensible claim than a latent-trait heritability. Parameters that could not be sourced independently are labelled scenario knobs and are never allowed to carry a headline.
3. **The headline is a boundary, not a number.** The other half of the problem — how the fertility *environment* moves, meaning how many children a person of fixed disposition has under prevailing conditions — is not measurable in advance and is where a projection can be made to say anything at all. Because selection and the environment both enter multiplicatively, they are directly commensurable, so we report the additional environmental decline that would exactly cancel measured selection (Section 6). That converts the free parameter into a threshold.

What we find

Measured mainstream selection raises world population in 2150 from 8.54 to 10.36 billion, a shift of 1.82 billion, and raises observed fertility 17.4% above the environment path. Adding the two named high-fertility groups with enough published demography to model separately — Haredi Jews in Israel and the Amish in the United States — adds only 0.05 billion more. The mechanism that popular discussion attaches to visible religious minorities is overwhelmingly a property of unlabelled variation inside ordinary populations.

The break-even environmental decline is 1.52% per decade after 2050 at the central measured parameters, with a 90% range of 1.51% to 1.53% across 50 paired stochastic migration paths, and a range of 0.33% to 4.49% per decade across the empirical uncertainty in the two measured parameters. Whether the world population in 2150 is nearer eight billion or nearer eleven depends on which side of roughly one and a half per cent per decade the real fertility environment falls, and we do not claim to know which.

What this is not

Three claims this paper does not make

It is not a forecast. The comparisons here are conditional on stated paths, and Section 6 exists precisely because the decisive quantity is unknown. We report where the model's answer changes sign, not what the answer is.

It is not a claim about genes. The transmission parameter is a parent–child correlation in realised family size. It combines whatever genetic, material and cultural continuity exists between generations, and no part of the argument requires separating them. Hruschka and Burger [2016] show that much of the observed variation in completed family size can arise from a counting process rather than from stable differences between women; Section 3 explains why the calibration used here survives that objection and what it concedes to it.

It is not an economic model. The fertility environment is a stated multiplier path, not a projection of income, housing or childcare costs. We describe its direction and report thresholds in it; we do not forecast it.

2 The mechanism

The model keeps two questions apart and answers them separately, because conflating them is what makes long-run fertility arguments irresolvable:

Who is becoming more common?
The composition of the population by fertility disposition. This is what selection moves.

How many children does a person of a given disposition have?
The fertility environment. This is what social and economic change moves.

Observed fertility is the product of the two. It is an output of the model, not an input to it, and that is the structural difference from a conventional projection, where observed fertility is the modelled object itself.

2.1 Types, propensities and transmission

Each country's population is divided along an extra axis into K fertility *types*. A type is not an ethnic, religious or political label; it is an effective fertility propensity ϕ_k, a multiplier on the country's own base-year age-specific fertility schedule. A woman of propensity 1.4 has forty per cent more children than the national average did in the base year, at every age, and that number does not change when conditions do. The propensities are normalised so that the base-year composition reproduces each country's published fertility exactly, which anchors the model to the present rather than fitting it to the future.

Writing ℓ for country, t for year, a for single year of age, k for type, $f_t[\ell, a]$ for the baseline age-specific fertility schedule and $E_t[\ell]$ for the environment multiplier, births are counted by the mother's type and then reassigned to the child's:

$$B_t[\ell, k] = \sum_a \bar{P}_t[\ell, k, \mathrm{F}, a] \; f_t[\ell, a] \; E_t[\ell] \; \phi[\ell, k], \tag{1}$$

$$\tilde{B}_t[\ell, j] = \sum_k B_t[\ell, k] \; M_t[\ell, k, j], \tag{2}$$

where $\bar{P}_t$ is mid-year exposure of women and M_t is the transition matrix from mother's type to child's type. Everything the mechanism claims is in those two lines. Survival, ageing, the open-ended age group and migration are untouched, and are carried identically for every type.

Selection is what happens when the rows of M differ and ϕ is not constant. Higher-propensity women appear more often in Equation (1) because they have more children; those children land in their mother's type more often than chance through Equation (2); so the composition drifts upward, so the population-average propensity rises, so observed fertility departs from the environment path without anyone imposing a floor on it. The quantity we report throughout is that departure,

$$S_t[\ell] \;=\; \frac{\text{observed fertility}}{\text{environment alone}} \;=\; \bar{\phi}_t[\ell], \tag{3}$$

the propensity averaged over women of childbearing age. It equals one when selection is switched off, and its distance above one is the size of the mechanism's claim.

2.2 The response, and why it is a covariance rather than a heritability

The mainstream transition rule is deliberately the simplest one that can carry the evidence. A child takes its mother's type with probability ρ and is otherwise drawn from the surrounding population of childbearing adults. Under that rule the one-generation response has a closed form. Mothers are sampled in proportion to their own fertility, so the birth-weighted distribution of types is $q_k \propto w_k \phi_k$ where w_k is the share of childbearing women of type k. The mean propensity of the newborn cohort is then

$$\rho \sum_k q_k \phi_k + (1-\rho) \sum_k w_k \phi_k \;=\; \bar{\phi} + \rho\, \frac{\sigma^2}{\bar{\phi}}, \tag{4}$$

so the relative gain per generation is $\rho\,\mathrm{CV}^2$, the transmission coefficient times the squared coefficient of variation of family size. This is the secondary theorem of natural selection in its covariance form [Robertson, 1966, Price, 1970]: the response equals the covariance of the trait with fitness divided by mean fitness, and here the trait and the fitness are the same quantity, so the covariance is a variance.

The important consequence is epistemic rather than algebraic. The right-hand side of Equation (4) is exactly Cov(parent family size, offspring family size) divided by mean family size, expressed relatively — and that covariance is directly observed. The model therefore moment-matches a measured intergenerational covariance. It does not require that any particular fraction of family-size variation be a stable inherited trait, which is the assumption that would be hard to defend and that Hruschka and Burger [2016] give good reasons to doubt.

At the central calibration ($\rho = 0.15$, $\mathrm{CV} = 0.57$) the one-generation response is 4.87%. Compounded over the roughly four generations between 2024 and 2150, that closed form implies a selection multiplier near 1.22, against the 1.174 the full projection produces. The closed form runs slightly hot because it ignores the age structure that delays each generation's effect and the migration that mixes compositions between countries; that the two agree to within a few per cent is a useful check that the simulation is doing what the theory says.

2.3 What children regress toward

Equation (4) contains a structural choice that turns out to matter more than almost anything else in the model, and that is easy to make by accident. When a child does *not* take its mother's type, which population does it resemble?

The current population.
The child is drawn from the composition alive around it. Selection then compounds, because the thing children regress toward is itself moving. This is the standard evolutionary-demography result and it is the specification used for the results below.

The base-year population.
The child is drawn from the 2024 composition, which never moves. Selection then runs into an equilibrium in which the pull-back cancels it, and the long-run effect is several times smaller.

Both are coherent. The second is a real claim — that institutions, schooling and media hold a society's fertility culture roughly in place regardless of who is in it — and it deserves to be argued for rather than arrived at silently. The first implementation of this model used the anchored rule without noticing, and produced a selection effect about a quarter of the theoretical size.

Because the two are kept as an explicit switch, the cost of the assumption is reportable: holding everything else fixed, anchoring transmission to the base-year composition lowers the selection multiplier from 1.174 to 1.072 and the 2150 world population from 7.61 to 7.13 billion. Nearly half a billion people at the horizon turn on a modelling decision about what a child regresses toward, which is a question about cultural transmission that demography could answer empirically and largely has not.

2.4 Named groups are transitions, not labels

Populations with both very high fertility and strong boundaries — Haredi Jews, the Amish, and others — are the version of this mechanism that reaches public discussion, and they are modelled here as one additional type with two properties rather than one. A named group has its own fertility, and it has its own *retention*: the share of children raised in it who remain in it as adults. Those who leave are not deleted into an average; they land in the mainstream, weighted toward its high-propensity type, on the reasoning that leaving a community is not the same as abandoning everything it taught you. How much they carry with them is a scenario knob (Section 3), set at 60% and reported separately for that reason.

Retention is as load-bearing as fertility, and the arithmetic makes this obvious: a group at one per cent of a population with six children per woman and ninety per cent retention, against a mainstream at 1.4, reaches 63% of the population in 4 generations, while the same group with weak retention goes nowhere. This is exactly the condition Arenberg et al. [2022] state — retention times fertility must exceed one for a group to grow — and because this model carries absolute fertility levels rather than heritability alone, the condition is evaluated rather than assumed. The projection's answer, given in Section 5, is that at world scale it barely matters. That is a finding, and it points away from the version of this story that gets told.

Two further limits are stated rather than fixed. No conversion *into* a named group is modelled, which is small for these two groups but not zero. And every type starts with its host country's age structure, because nobody publishes one by type; since a population with six children per woman is far younger than its host, this understates group growth, and every group share reported below is a lower bound.

3 Calibration

Of the mechanism's 13 parameters, 8 are estimated from published evidence and 5 are not. The distinction is enforced in code rather than described in prose: the parameter loader refuses a row with no provenance, refuses a row marked as sourced that carries no evidence, and refuses a scenario knob that claims to be verified. The full table with sources is Appendix A.

One rule governs the whole exercise, and it is the reason the numbers in Section 5 mean anything:

> **The standing rule on parameters**
> No mechanism parameter may be fitted to the series it is meant to explain. If a quantity cannot be sourced independently of the fertility path the model produces, it is labelled a scenario knob and is never allowed to carry a headline result. This is a restriction on what the model may learn, adopted because a mechanism tuned to reproduce observed fertility explains nothing about it.

3.1 The dispersion of completed family size

The first mainstream parameter is the coefficient of variation of completed family size. It comes from the Cohort Fertility and Education database [Wittgenstein Centre, 2017], which reports parity distributions by birth cohort. We pin 43 country files, and apply one declared rule everywhere: women rather than couples or mothers only; children ever born, including childless women; whole birth-cohort bins whose members were all aged 50–59 at observation; and one estimate per country, with education categories summed.

The open highest-parity category has to be handled, and it does not have to be guessed at, because the database supplies exact total children and therefore identifies the category's mean. The primary estimate distributes it geometrically, the maximum-entropy integer distribution with that mean. Substituting the minimum-variance integer tail instead moves the coefficient of variation by 0.005 at the median across all 43 countries and by at most 0.059.

Across the 19 countries with mean completed fertility at or below 2.2 children (29.1 million women), the country-median coefficient of variation is 0.570, with an observed range of 0.443 to 0.787. We take the unweighted country median rather than the women-weighted one because these sources are a set of plausible settings rather than a harmonised global sample, and weighting would let the single largest national extract dominate. The adopted value is 0.57 with a propagated range of 0.44 to 0.80.

As an independent check on a source with a different construction, paired cohort fertility tables from the United States National Center for Health Statistics [National Center for Health Statistics, 2024] give mean completed fertility 2.04 and a coefficient of variation of 0.695 for cohorts born 1947–1956 at exact age 50. That is inside the declared range and above the cross-country median.

3.2 What that dispersion is, and is not

Hruschka and Burger [2016] show, using two hundred surveys, that a large share of completed-fertility variance can be produced by a counting process alone, without any stable differences between women. Their objection is correct and it applies here. The parity marginals used above do not identify a latent trait with coefficient of variation 0.57: realised family size also contains timing, infertility, partnership history and measurement error.

Two things follow. First, the parameter is used as an *effective phenotypic spread*, paired with the observed intergenerational correlation, and the pair reproduces a measured covariance — Section 2.2 — rather than asserting a variance decomposition. Second, the correction that would seem natural is unavailable: most of these low-fertility distributions are *under*-dispersed relative to Poisson, consistent with targeted family sizes, so subtracting a Poisson variance to manufacture a latent-trait coefficient of variation would be invalid. That is why the empirical range 0.44–0.80 is carried through every result rather than collapsed to a point.

3.3 Parent–child transmission

The second mainstream parameter is the parent–child correlation in completed family size. Murphy [1999] reports correlations mostly between 0.15 and 0.20 in the larger recent British and American samples reviewed there. Murphy [2013], covering 46 populations in 28 countries, reports unweighted regional means for respondents aged 40 and over of 0.09 in Northern America, 0.11 in Northern Europe, 0.15 in Western Europe, 0.17 in Southern Europe and 0.19 in Eastern Europe. We adopt 0.15, with a deliberately conservative cross-setting interval of 0.05 to 0.30.

This is the whole of the transmission assumption, and it is worth being explicit about what it excludes. Twin studies report substantial genetic variance components for fertility behaviour [Kohler et al., 1999], and there is direct molecular evidence of contemporary selection on fertility-associated variants [Beauchamp, 2016, Kong et al., 2017]. None of that is used. A heritability is a variance decomposition and is not numerically interchangeable with a parent–child correlation; substituting one for the other would inflate the mechanism by a factor of two or more with no warrant. The parameter used here is the observed continuity, whatever produces it.

3.4 The named groups

Two populations have enough published demography to pin fertility, retention and population share separately, which is the threshold for modelling a group by name rather than inventing one.

Haredi Jews in Israel: fertility 6.40 children per woman, population share 13.9%, and retention 86.7% implied by an estimated 13.3% of adults raised Haredi having left [Regev and Gordon, 2020, Israel Democracy Institute, 2024]. The retention interval is wide (0.735–0.950) because the same study documents leaving rates from 5.4% to 26.4% across factions, and that heterogeneity is real rather than sampling noise.

The Amish in the United States: fertility 6.10 children per woman and retention 84.5% at ages 40 and over, from a population-wide analysis of more than 50,000 households [Anderson and Thiehoff, 2025], with a population share of 0.115% from a specialist census of settlements [Young Center, 2024].

Both fertility figures are period rates used only to initialise a relative propensity at the model's base year. They are not used as scoring targets, since the model's claim concerns cohort fertility and period rates are contaminated by timing [Bongaarts and Feeney, 1998].

3.5 The parameters that are not measured

5 parameters cannot be sourced independently and are labelled scenario knobs. The number of mainstream types (3) is a discretisation choice. The share of group leavers who land in the mainstream's high-propensity type (60%) has no direct measurement. The rate at which a named group's fertility advantage decays as the group stops being unusual has no established general value. And two describe the fertility environment: the additional proportional decline per decade, and the floor beneath it.

The last of these is the important one, and it is the quantity this project exists to be uncertain about. It cannot be looked up, and it must not be fitted, because fitting it to observed fertility would use up exactly the evidence the mechanism is supposed to explain. Section 6 is the response to that problem: rather than choosing a value and reporting the population it implies, we report the value at which the model's answer changes sign.

4 The projection engine, and why it is worth validating first

A mechanism is only worth reporting if a reader can tell the difference between its effect and an arithmetic error. That requirement drove the order of the work: the accounting layer was built and validated against published outputs before any mechanism was added to it, and the mechanism was then implemented as an extra axis on the same code rather than as a new model.

4.1 The accounting layer

The engine is a standard cohort-component projection with no theory in it. It is single year of age from 0 to 100 with an open-ended final group, two sexes, 237 countries and territories, annual steps, populations dated 1 January and rates in force during the calendar year. It takes fertility, survival and net migration as given and does the bookkeeping. Its inputs come from World Population Prospects 2024 [United Nations Population Division, 2024a]: age-specific fertility, survival ratios into each age, and sex ratio at birth.

Two conventions are worth stating because getting them wrong produces plausible rather than obviously broken output. Survival $s[a]$ is survival *into* age a, which is the UN's own convention, so that the numbers in the engine are the numbers in the source file with no translation step. And childbearing ages run from 10 to 54 rather than 15 to 49: the UN's single-age fertility file stops at 49, its five-year file does not, and the missing mothers are about 0.3% of world births — small in one year, and a visible bias by 2100.

Check at 2100, WPP 2024 zero-migration variant	Result	Tolerance
World population	0.001%	0.05%
Worst country above 10,000 people (COK)	0.13%	0.5%
Worst five-year age group below 100	0.006%	0.05%
Open-ended 100+ group	1.08%	2%
Constant fertility vs the UN's own variant	0.05%	0.5%

Table 1: The engine reproduces the United Nations' published projection from the United Nations' published inputs. The zero-migration variant is the real test because nothing in it is estimated by us. The medium variant agrees to 0.04% but is reported as a diagnostic rather than a test, because the UN does not publish net migrants by single year of age and ours are backed out of its own medium path as a residual.

4.2 The validation

The test that matters is the zero-migration variant, because every input to it is published and nothing in it is tuned, so any discrepancy is our arithmetic being wrong. Running from 1 January 2024 to 1 January 2100 on the UN's own inputs (Table 1), world population lands within 0.001% of the UN's published figure, the worst of 227 countries above ten thousand people is within 0.13%, and the worst five-year age group below 100 is within 0.006%.

The constant-fertility comparison is the absurdity check at the other end of the range: freezing the base year's rates gives about 53 billion people by 2150, within 0.05% of the UN's own constant-fertility variant at 2100. Worth recording is that an earlier specification of this project expected roughly 244 billion, a figure taken from a UN long-range report built on the 2002 revision. Rates have fallen a great deal since 2002, so reaching 244 billion from a 2024 base would have meant the engine was wrong. A remembered number was treated as a requirement for longer than it should have been.

4.3 Why the mechanism's effect is attributable

The typed engine of Section 2 is the same code with a composition axis. Setting every propensity to one leaves the composition free to move but unable to change fertility, and in that configuration the typed engine reproduces the ordinary engine to a relative difference of 3×10^{-16} — machine precision. It also lands on the same 2150 world total as the older single-axis long-run diagnostic.

This is not a formality. It means the difference between a run with selection and a run without it cannot be a second implementation of the arithmetic, a different handling of the open age group, or a changed exposure convention. It is the mechanism. Separately, one generation of the typed engine matches the analytic parent–offspring covariance response of Equation (4) to nine decimal places, which is the implementation agreeing with its own calibration.

4.4 Migration, and what it is allowed to support

The UN does not publish net migrants by single year of age. For the runs in Section 5 the migration input is backed out of the UN's own medium path as a residual, which makes it a usable forward-model input and no evidence at all: it absorbs every difference between the UN's procedure and ours. Runs using it are labelled diagnostics.

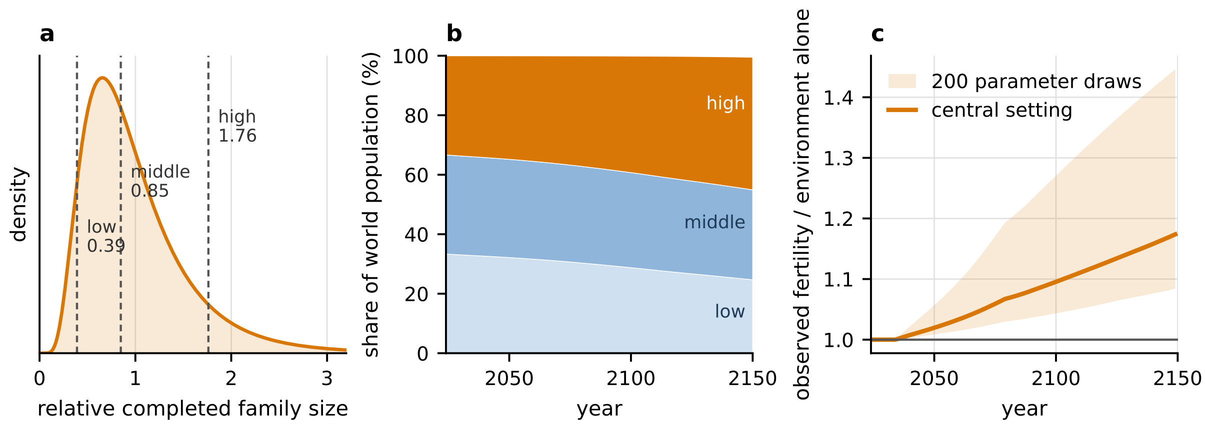

Figure 1: The mechanism in three steps. **(a)** Completed family size is right-skewed with a coefficient of variation of 0.57; the model stands three effective types in for that distribution, at the conditional means of its equal-probability thirds, rescaled to reproduce the dispersion exactly. **(b)** World composition drifts because higher-propensity women contribute more of the next cohort: the high type rises from 33.3% of the world population at 2024 to 44.6% at 2150. **(c)** The resulting selection multiplier, observed fertility divided by the environment path alone, with the band showing 200 draws from the stated ranges of all 13 mechanism parameters.

Where migration uncertainty is the object of study — Section 7 and the robustness check in Section 6 — a different and stronger construction is used: the published Bayesian migration trajectories of Azose and Raftery [2015], with rates applied to each path's own evolving population and every draw-year balanced so that world net migration is exactly zero. That constraint matters more than it sounds. The public trajectory archive is balanced in expectation but not draw by draw, and using it unbalanced allows individual paths to create or delete millions of people globally. An earlier version of the uncertainty decomposition reported below did exactly that and attributed 1.75 billion of world spread to migration; a component that cancels globally by construction cannot do that, and the corrected figure is 0.34 billion.

5 How large is selection?

5.1 The size of the effect, step by step

The cleanest benchmark is not the UN's own path, because that path already contains an assumed fertility recovery whose justification is exactly what is in dispute. Instead the benchmark holds that today's fertility already embodies today's conditions: the decline already visible in the data continues to its observed floor and then stops, with no assumed rebound and no assumed further deterioration. Selection is then switched on against that background, and the components are added one at a time (Figure 2).

Measured mainstream selection — the two independently sourced parameters, with no named groups and no assumed change in conditions — raises world population in 2150 from 8.54 billion to 10.36 billion. That is 1.82 billion, or 21%, from a mechanism that is entirely absent from conventional projections. Observed world fertility ends 17.4% above the environment path that produced it, and rises above it detectably from about 2042 (Figure 1c).

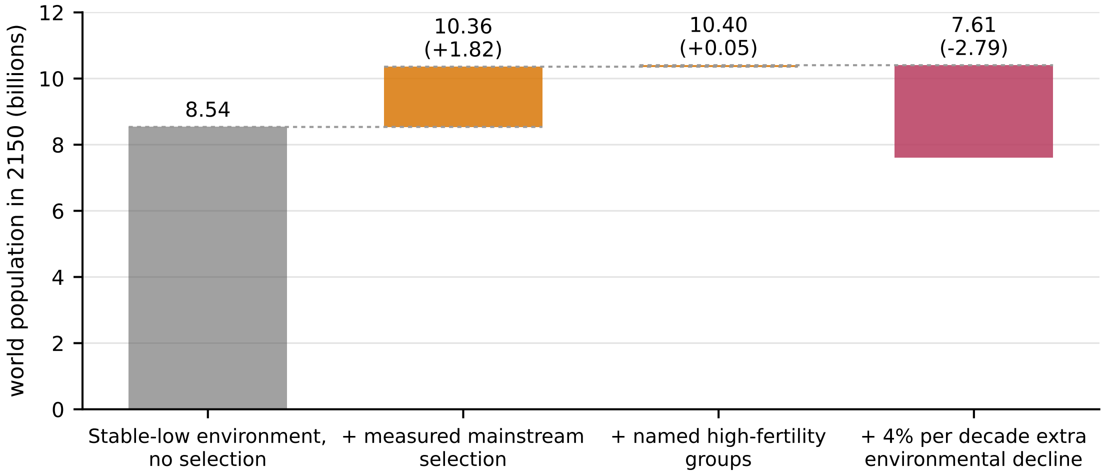

Figure 2: World population in 2150 as each component is added. Grey is the benchmark environment with no selection. Orange steps are the mechanism: measured mainstream selection first, then the two named high-fertility groups. Red is the illustrative stress test, a further 4% per decade decline in the fertility environment after 2050, which is a scenario knob and not a forecast. The 2.79-billion size of that last step, against 1.82 billion for the mechanism itself, is why this paper's primary result is the boundary in Section 6 rather than any single number on this chart.

The mechanism is a level shift, not a return to growth. Under the benchmark environment with mainstream selection the world still peaks and then runs close to flat; selection changes where the curve settles, not its shape. This is the distinction Arenberg et al. [2022] press against models built from heritability alone, and it survives here because the projection carries absolute age-specific fertility rather than a transmission coefficient by itself.

5.2 Named groups are a small part of the story

Adding the two named high-fertility groups — with their measured fertility, their measured retention, and one scenario knob routing their leavers — moves the world result by 0.05 billion. The world composition at 2150 is 0.5% named-group.

This deserves emphasis because it inverts the usual telling. Public discussion of selection on fertility is almost entirely about visible high-fertility minorities, and this model treats them generously: it gives them their measured fertility, it lets high retention compound for four generations, and its assumption that each group starts with its host country's age structure understates their growth, so the shares reported here are lower bounds. They still contribute 2.5% of what unlabelled variation inside ordinary populations contributes. The reason is arithmetic rather than sociological. A group at a tenth of a per cent of a national population has four generations to compound before it is large enough to matter to a world total, whereas the mainstream's dispersion acts on everybody at once from the first year.

Where a group is already large it does matter, and locally it dominates. In Israel, whose Haredi population is 13.9% of the country at the base year, the named group reaches 58% of the population by 2150 and the country's selection multiplier reaches 1.81 — half again as much fertility as the environment path alone would give. In the United States the Amish reach 4.6%, contributing

Country	Selection multiplier at 2150	Composition of the effect
Israel	1.81	mainstream plus a large named group
United States	1.30	mainstream plus a small named group
Nigeria	1.16	mainstream only
Japan	1.15	mainstream only
Germany	1.15	mainstream only
Korea	1.14	mainstream only

Table 2: Observed fertility divided by the environment path alone, at 2150, in the run that includes both selection and the 4% stress test. The multiplier is a statement about composition and is unaffected by the level of the environment path, for the reason given in Section 6.1.

to a national multiplier of 1.30. Neither of these is a world-scale effect, and neither should be reported as one.

5.3 Where selection bites, and where it does not

The same run produces very different multipliers by country (Table 2), and the ordering is informative. Selection is strongest where fertility is already low, because the model's response depends on the *relative* dispersion of family size rather than its absolute level: a country at 0.8 children per woman has as much proportional room for recomposition as one at 4, and it passes through more generations in the same number of years.

5.4 The mechanism's own uncertainty

Drawing all 13 mechanism parameters from their stated ranges 200 times gives a selection multiplier at 2150 between 1.084 and 1.446 at 90%, and a world population in the stress-test scenario between 5.84 and 11.56 billion. Selection overtakes the environment in 100% of those draws.

The direction is therefore robust and the magnitude is not, and it is worth being precise about which parameter is responsible. Almost all of the width comes from the empirical dispersion of family size: holding everything else fixed, the endpoints of the measured range, 0.44 and 0.80, move the stress-test world result from 7.10 to 8.92 billion. That is a genuine empirical uncertainty in a quantity that is measurable, unlike the environment path, and narrowing it is tractable work: it needs completed-cohort parity distributions from more than the small set of countries that publish them.

6 The boundary

Figure 2 contains an uncomfortable fact. The mechanism, built from two measured parameters, moves the 2150 world result by 1.82 billion. A single unmeasured scenario knob — how fast the fertility environment keeps falling — moves it by 2.79 billion in the other direction. If that knob were reported as a preferred value, it would be doing most of the work while carrying none of the evidence, and the paper would amount to an assertion dressed as a projection.

There is a better use for it, available because of a structural property of the model.

6.1 Selection and the environment are commensurable

The environment multiplier $E_t[\ell]$ in Equation (1) multiplies every type equally. It therefore cancels out of the relative weights with which types contribute to the next generation: whether conditions are generous or harsh, the ratio of a high-propensity woman's births to a low-propensity woman's births is unchanged, so the composition follows almost the same path either way.

The cancellation is exact in the birth weights and very nearly exact in the outcome. Running the same full selection setting under two environments that end 2.79 billion people apart gives selection multipliers of 1.176 and 1.174, a difference of 0.22%. The residue is not a rounding error but a real and small second-order effect: a harsher environment produces a different age structure, so the childbearing-age women over whom the propensity is averaged are drawn differently, and a fixed number of migrants mixes into a smaller population with proportionally more force. Both effects are worth naming because a model in which they were exactly zero would be one that had lost track of the population's age structure.

That near-separability is a modelling assumption, and Section 8.3 takes it seriously as one. But it is also what makes the two forces directly comparable. Selection multiplies fertility by S; a uniform environmental decline of rate r per decade over d decades multiplies it by $(1-r)^d$. The two cancel when

$$S \cdot (1-r)^d = 1 \qquad \Longleftrightarrow \qquad r^\star = 1 - S^{-1/d}. \tag{5}$$

So instead of asking what the population will be, we can ask a question whose answer is determined by measured quantities: *how much additional environmental decline would exactly cancel measured selection?*

6.2 The break-even decline

At the central calibration, measured mainstream selection reaches $S = 1.165$ in the final fertility-rate year (2149), giving a break-even rate of 1.53% per decade of additional decline beginning in 2050. Country selection effects are aggregated using births from the no-selection counterfactual, so a scenario cannot choose the weights by which it is judged.

Figure 3 shows how that threshold moves across the empirical range of both measured parameters, over a 33-cell grid. It runs from 0.33% per decade at the low corner — weak dispersion and weak transmission — to 4.49% per decade at the high corner. The threshold is close to linear in the transmission coefficient and roughly quadratic in the dispersion, as Equation (4) implies it should be.

Read as a statement about the world, Figure 3 says something specific and, we think, more useful than a projected total. If the conditions that determine how many children a given person has continue to deteriorate at less than roughly one and a half per cent per decade after 2050, the compositional force wins and world fertility ends the century-and-a-half above where the demographic transition left it. If they deteriorate faster than that, the environment wins. Nobody can currently say which, and this paper does not try. What it does is put a number on the question, so that a claim about the future of fertility can be stated as a rate that could in principle be observed rather than as a mood about modernity.

For scale: 4% per decade, the illustrative stress test, is about two and a half times the break-even rate and is far from a neutral choice. Sustained for a century, it removes a third of the fertility

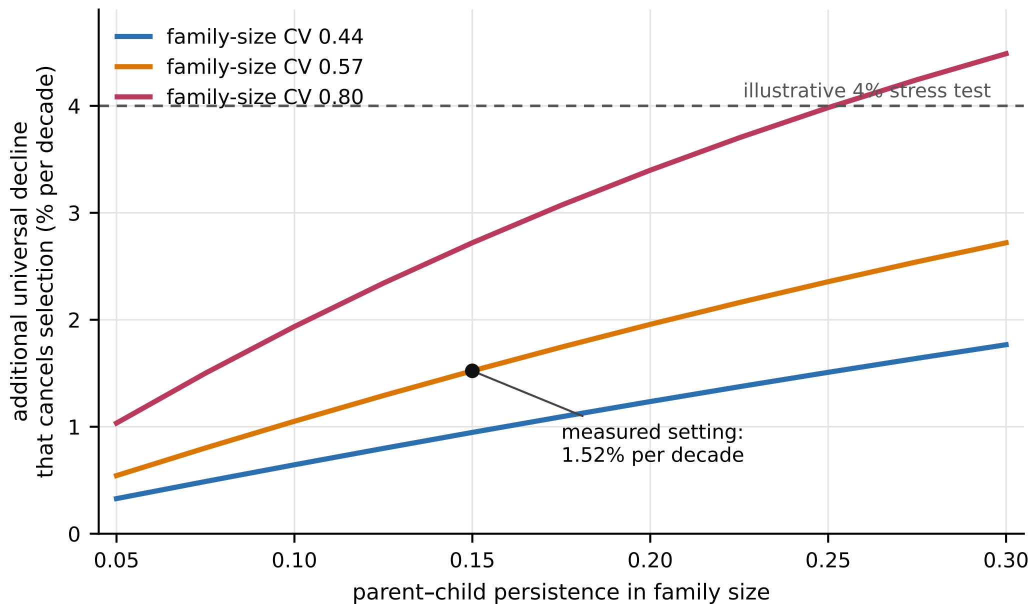

Figure 3: The additional uniform decline in the fertility environment, per decade after 2050, that exactly cancels mainstream selection by 2150. Above a curve, the environment wins and fertility ends below the benchmark; below it, selection wins. The two axes are the measured parameters: the dispersion of completed family size (three curves, spanning the observed low-fertility country range) and the parent–child correlation in it. Shaded bands are the 5th to 95th percentile across 50 paired stochastic migration paths and are narrow enough to be nearly invisible, which is itself the result reported in Section 6.3. The dashed line marks the illustrative 4% stress test used in Figure 2.

environment before its floor binds. We report it because it was the project's original central scenario and because a stress test should be visible, not because we think it likely.

6.3 The boundary survives stochastic migration

A boundary computed on one migration path would be worth little if migration could move it. It is checked against 50 stochastic paths drawn from the published Bayesian migration trajectories [Azose and Raftery, 2015], with one important design choice: the selection run and its no-selection counterfactual share each source path and each post-2100 innovation, so the comparison is paired and the migration draw cannot masquerade as a selection effect. Every draw-year is balanced to exactly zero world net migration.

The central boundary is 1.52% per decade with a 90% range of 1.51% to 1.53%. Migration moves the threshold by about one part in seventy-five, against a factor of thirteen between the corners of the measured-parameter grid. This is what one should expect — migration sums to zero globally by construction and can only affect a world fertility aggregate by relocating people between countries with different schedules — but expecting it is not the same as having checked it.

6.4 What the boundary is not

Reading Figure 3 correctly

The vertical axis is *additional* decline, on top of the fertility fall already present in the observed data and already carried by the benchmark path. It is not a forecast of economic conditions, and this paper makes no claim about what causes the fertility environment to move or how fast it will. The threshold is also a statement about the terminal year: because both forces compound, a decline that begins later or ends earlier trades against the same total. And the uniformity is deliberate — a decline that fell unequally on different dispositions would interact with selection rather than merely oppose it, which is a richer model than the evidence currently supports and would destroy the clean threshold that makes this figure readable.

7 The horizon, and what is whose

A projection that runs to 2150 using inputs that stop at 2100 has a problem of attribution rather than of arithmetic. Everything past the last year of published assumptions is the modeller's own extrapolation, and if it is drawn as one continuous line the reader cannot tell where the source ended. This section states the boundary explicitly and reports what crosses it.

7.1 Three separate objects

The UN reproduction.
World Population Prospects 2024's medium path, reproduced by this engine, through 2100 and no further. No value after that year is labelled a UN projection.

The project extension.
Begins from the published population on 1 January 2100 and runs to 2150. Fertility, mortality and the sex ratio at birth are held at their final published age schedules, which is our assumption and not the source's. Migration continues stochastically.

The selection model.
A separate object that forks in 2024, so that selection acts over the whole projection rather than being switched on after the UN's assumptions have already determined the 2100 population.

7.2 The extension, and why its band is narrow

The published Bayesian migration trajectories [Azose and Raftery, 2015] run to 2100 and the public archive does not include the fitted posterior state, so the continuation refits the same autoregressive form to the 2070–2100 portion of those trajectories and continues each path from its own 2100 rate. This is an emulator of a model's output, not an official continuation, and it is labelled as one throughout.

The result is 10.187 billion in 2100, 9.573 billion in 2125, and 8.725 billion in 2150, with a migration-only 90% range of 8.656 to 8.772 billion. The narrowness of that band is a structural fact rather than a claim of precision: net migration sums to zero across the world in every draw-year, so it can only move a world total indirectly, by relocating people into countries whose fertility and mortality schedules differ from the ones they left.

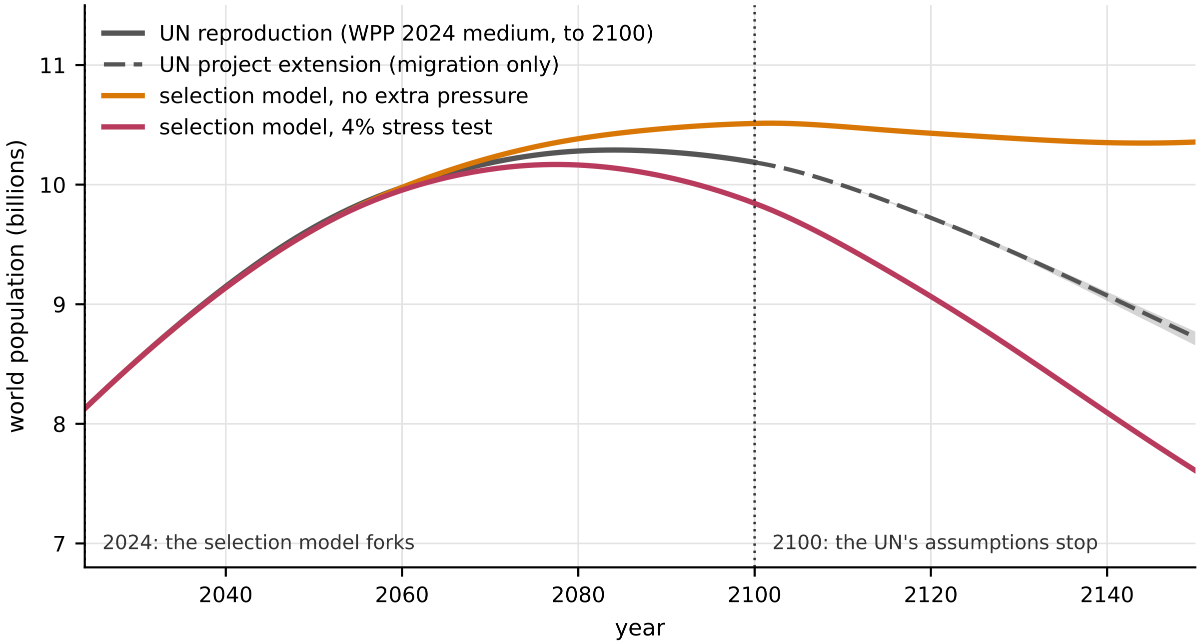

Figure 4: Three objects, drawn so that it is visible which is whose. The solid grey line is the reproduced UN medium path and stops where the UN's assumptions stop. The dashed line and shaded fan are this project's own continuation, in which fertility and mortality schedules are frozen and migration varies across 1000 paths; the fan is narrow because migration cancels globally by construction. The two coloured lines are the selection model, which forks in 2024 and is not a UN projection at any point.

7.3 Which uncertainty dominates depends on where you look

Varying one source at a time across 200 draws (Figure 5a), the 90% width of world population in 2150 is 7.26 billion for long-run fertility, 5.72 billion for the selection mechanism's own parameters, 0.73 billion for our post-2100 hold-constant rule, 0.52 billion for mortality and 0.34 billion for migration. The mechanism entry is not directly additive with the others, because it varies mechanism parameters against the UN's medium demographic rates rather than against the probabilistic posterior; it is included because a decomposition that omitted the paper's own subject would be a strange thing to publish.

Two of these deserve comment. Our own post-2100 rule contributes more world spread than mortality and migration combined, which is a useful corrective: at this horizon the modeller's housekeeping choices are not negligible relative to the components everyone argues about. And the selection mechanism is second only to long-run fertility, which is the case for treating it as a first-class source of uncertainty rather than an omitted refinement.

The country panel (Figure 5b) disagrees with the world panel, and that disagreement is the point. At 2100, migration uncertainty is 42.16 times fertility uncertainty for the United Arab Emirates and 3.99 times for Canada, but only 0.06 times for Nigeria. In the project extension the Emirates' 2150 population ranges from 1.0 million to 267.3 million around a median of 24.6 million — a factor of more than two hundred, for a component that contributes almost nothing to the world total. Which uncertainty dominates is a fact about where you are looking, and any single summary

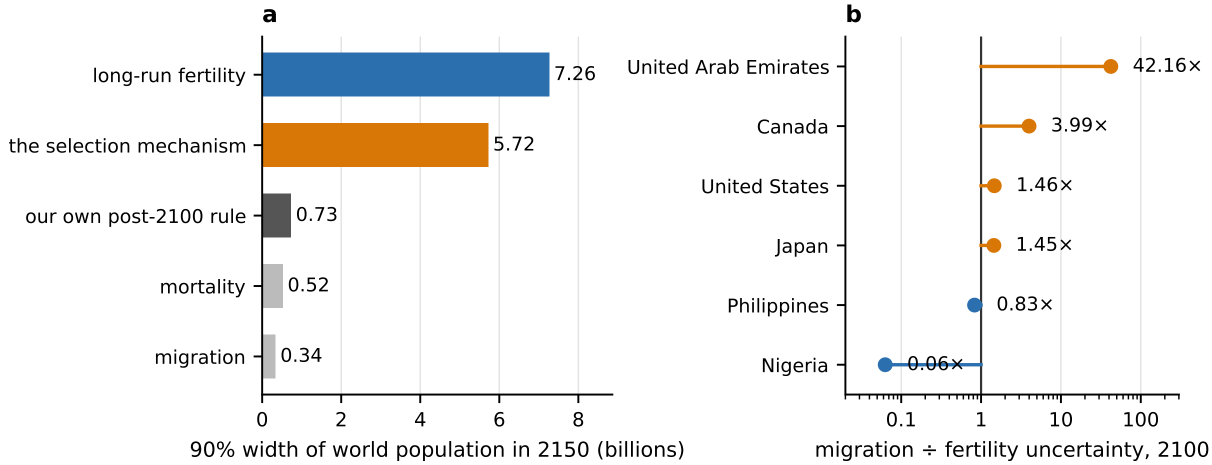

Figure 5: **(a)** The 90% width of world population in 2150 attributable to each source, varying one at a time across its own draws with the others held at a median trajectory. The post-2100 entry is what our own hold-constant rule *adds*: it is the extra width from ending the source data ten years earlier, which is the cleanest available measure of how much of the answer is ours rather than the source's. **(b)** The same comparison inside individual countries at 2100, where the ordering reverses.

of "the" uncertainty in a population projection is answering a question about the world that most users of projections are not asking.

7.4 A conventional comparator, recorded in advance

For a benchmark that nobody could accuse us of arranging afterwards, the published Bayesian posterior for fertility and mortality [Alkema et al., 2011, Raftery et al., 2012, Ševčíková and Raftery, 2016] was propagated through this same engine, one draw at a time: 1000 draws, 236 countries, to 2150. Its median peaks at 10.31 billion in 2093 and reaches 9.73 billion in 2150, with a 90% band of 6.97 to 14.36 billion.

This is the conventional mean-reverting view, not this paper's. It was computed, stored as an immutable dated record, and marked as not a project claim *before* the mechanism results existed, precisely so that the comparison could not be assembled to flatter the mechanism afterwards. One check on it is worth reporting because nobody arranged it: the wholly separate deterministic run on the UN's own published assumptions peaks at 10.29 billion in 2084, against the ensemble median's 10.31 billion in 2093. Those two calculations share only the engine, and they agree on the peak to within a fifth of a billion people and on its date to within a decade.

7.5 What the historical record says about published ranges

The same archive machinery grades 8 archived UN revisions from 1992 to 2008 against the current estimates. World population was under-projected by -2.45% on average and in the same direction in every revision since 1996; African fertility was projected -9.8% too low and East Asian fertility 14.9% too high; life expectancy was -1.32 years too low.

We report this not as evidence about our own model, which it is not, and not as a reason to prefer higher numbers. Its useful content is about ranges rather than central values: only 41 of 117 world-level projections (35%) fell inside the low-to-high band printed in the revision that made them. Those bands were fertility scenarios without probabilities attached, so this is not a calibration failure in the technical sense. It is a demonstration that the intervals demography has historically published around long-run population have been too narrow, which is the case for the kind of explicit uncertainty accounting attempted here.

8 Discussion

8.1 A middle position, and why it is a position rather than a hedge

Two literatures reach opposite conclusions about what intergenerational transmission of fertility implies. Collins and Page [2019] apply the breeder's equation to national fertility and conclude that global population stabilisation this century becomes very unlikely; the European and North American trajectories in their model turn from decline to growth. Arenberg et al. [2022] reply that a model built from transmission alone contains no absolute fertility level, cannot represent the fact that higher-fertility subgroups are themselves converging downward, and therefore cannot establish that transmission produces growth at all.

Both are right about what they are arguing. The disagreement is not really about the mechanism, which nobody disputes; it is about whether a model that carries only a transmission coefficient can support a conclusion about population size. It cannot, and the fix is to put the mechanism inside a projection that carries levels.

Doing that yields a result that neither side would predict from the other's setup. The mechanism is real and measurable, its direction is unambiguous, and it is worth 1.82 billion people at 2150 — far too large to leave out of a long-run projection. It is also not a return to growth. Under the benchmark environment with mainstream selection, the world still peaks and then runs close to flat; what selection changes is where the curve settles. The Arenberg condition is satisfied rather than violated, because absolute age-specific fertility is carried at every step, and the outcome is a level shift of about 21%.

The second finding is more surprising and cuts against the way this subject is usually discussed. Explicitly named high-fertility groups contribute 2.5% of what unlabelled variation contributes, even though the model is generous to them at every choice point. The compositional force in long-run fertility is not a story about visible minorities; it is a property of the ordinary dispersion of family size, acting on everybody at once. That is a better result than the alternative, because ordinary dispersion is a quantity that national statistical offices already measure.

8.2 On the environment, and why we report a threshold

The honest position on the fertility environment is that nobody knows what it will do. The literature contains coherent arguments in both directions: the low-fertility trap hypothesis [Lutz et al., 2006] describes self-reinforcing downward pressure through ideal family size and cohort experience, while the mean-reverting specifications used in official projections [Alkema et al., 2011, Raftery et al., 2012] embed an eventual recovery. Neither is settled, and the sensitivity of a 2150 world total to the difference is enormous — 2.79 billion for the illustrative path used here.

A projection whose headline is controlled by that quantity is reporting its author's intuition with extra steps. Equation (5) is the alternative: because both forces multiply fertility, the model can be asked for the environmental decline that exactly offsets the measured compositional gain, and that number — 1.52% per decade after 2050, robust to migration to within 1.51%–1.53% — is determined by the two measured parameters and the arithmetic.

We would defend this as the right output for a problem of this shape. It is falsifiable in the parameters that are measurable, it is explicit about the one that is not, and it turns a disagreement about the future into a disagreement about a rate. Someone who believes the fertility environment will keep deteriorating rapidly and someone who believes it has nearly bottomed out can now locate their disagreement on Figure 3 rather than trading projections.

8.3 Limitations

The environment multiplies every type equally.
This is what makes the boundary a single clean number, and it is the assumption most likely to be wrong. Rising costs of housing, childcare and foregone earnings plausibly fall hardest on exactly the people whose disposition would otherwise produce large families, in which case the environment would interact with selection rather than merely oppose it and no single threshold would exist. Modelling that interaction requires evidence on how fertility declines have differed across the family-size distribution within cohorts, which we did not find at the coverage needed. We regard this as the most valuable single extension.

Three types are a discretisation, not a discovery.
The propensity distribution is stood in for by 3 points chosen to reproduce its first two moments exactly. Higher moments, and the shape of the upper tail, are structural approximations. The number of types is a declared scenario knob for that reason.

The dispersion parameter is an effective spread, not a latent trait.
The concession to Hruschka and Burger [2016] in Section 3 is genuine. What is defended is the moment match to an observed covariance; what is not claimed is that the observed variance in completed family size is inherited. Because the covariance is what enters Equation (4), this is sufficient for the one-generation response, and it is the extrapolation across four generations that the empirical range 0.44–0.80 is carried to reflect.

Transmission is one constant, applied everywhere.
A single parent–child correlation is used for all 237 countries, when the underlying literature reports regional means ranging from 0.09 to 0.19 and does not cover most of the world. Countries outside Europe and North America are effectively assigned a parameter estimated inside them.

Named groups are modelled without conversion in, and with their host's age structure.
Both push the same way, so the reported group shares are lower bounds, but neither is small in principle and a group large enough to matter would deserve its own age structure.

After 2100 the demographic schedules are held constant.
This is our rule and not the source's, and Section 7 measures what it is worth (0.73 billion of world spread at 2150) rather than leaving it implicit.

Migration by age is borrowed.
The United Nations does not publish net migrants by single year of age. Where migration is an input it is a residual backed out of the UN's own path, which is usable but is not independent

evidence; where migration is the object of study its level comes from published trajectories but its age and sex composition is still borrowed.

Selection is on fertility only.

Survival is identical across types. High-fertility groups often have distinctive mortality, and modelling that without evidence would add a parameter and subtract credibility.

8.4 What would show this to be wrong

The mechanism's claim resolves in cohort fertility, not period fertility, and the distinction is not pedantry: period rates are depressed by postponement even when completed family sizes are unchanged [Bongaarts and Feeney, 1998], so a mechanism predicting a rise in completed cohort fertility could be scored as failing for decades by a measure that is not measuring it.

The implication is uncomfortable but should be stated: the first genuinely informative test of this model arrives when the cohorts born around 1990 complete their childbearing, in the late 2030s. What can be checked before then is narrower — whether the dispersion of completed family size behaves as Section 3.1 measures it in countries not yet in the sample, whether the parent–child correlation holds up outside Europe and North America, and whether the composition drift in Figure 1b is visible in successive cohorts of the same country.

Because a claim that resolves in 2038 is easy to revise quietly in 2035, the projections behind this paper are written as immutable dated records with proper scoring rules attached [Gneiting and Raftery, 2007], and the storage layer raises rather than overwrite an existing record. The comparator of Section 7.4 was deposited before the mechanism results existed, and is marked as not a claim of this project. We think this record-keeping, rather than any particular number in Section 5, is the part of the work most likely to still be useful in thirty years.

9 Conclusion

Long-run population projection is dominated by a single quantity nobody can measure: what fertility does after the demographic transition. Conventional practice handles this by modelling the national rate as a mean-reverting time series, which is a defensible statistical choice and an odd substantive one, because it assumes away the fact that the parents of each generation are sampled in proportion to how many children they have.

Adding that compositional force to a validated cohort-component projection shows it to be real, measurable from evidence independent of the series it explains, and consequential: 1.82 billion people at 2150, or 21%, from two parameters that statistical offices already have the data to estimate. It also shows it to be more modest than either its advocates or its critics would suggest. It shifts the level of the long-run curve; it does not restore growth; and it is almost entirely a property of ordinary variation rather than of the visible high-fertility minorities that dominate discussion of the subject.

Against that measured force sits an unmeasurable one, and the paper's main contribution is a way of reporting the comparison that does not require guessing. The two act multiplicatively on fertility, so the question "what will the population be?" can be replaced by "how fast would conditions have to keep deteriorating for the compositional force to lose?" The answer, at the central measured calibration, is 1.52% per decade after 2050, and it ranges from 0.33% to 4.49% across the empirical uncertainty in the two measured parameters.

We do not know which side of that threshold the world is on. We would rather report the threshold than pretend otherwise, and we would rather be wrong for a reason that can be located on Figure 3 than be approximately right for none.

A Mechanism parameters

Table 3 is generated directly from the machine-readable table the model loads, so it cannot drift from the values the results were produced with. Each row carries an evidence field and a provenance field in the source file; the loader refuses to run if a row marked `sourced` carries no evidence, or if a row marked `knob` claims to have been verified.

Parameter	Kind	Value	Range	Unit
mainstream persistence	sourced	0.150	0.050–0.300	correlation
mainstream propensity cv	sourced	0.570	0.440–0.800	coefficient of variation
group retention haredi	sourced	0.867	0.735–0.950	proportion
group fertility haredi	sourced	6.400	5.800–7.000	children per woman
group share haredi	sourced	0.139	0.120–0.150	proportion of population
group retention amish	sourced	0.845	0.800–0.920	proportion
group fertility amish	sourced	6.100	5.000–6.500	children per woman
group share amish	sourced	0.00115	0.00110–0.00130	proportion of population
mainstream types	knob	3	—	count
defector high propensity weight	knob	0.600	0.300–0.900	proportion
development decline per decade	knob	0.040	0.000–0.080	proportional fall
development floor	knob	0.550	0.400–0.750	multiplier
group convergence per generation	knob	0.100	0.000–0.250	proportional fall

Table 3: All 13 mechanism parameters. Of these, 8 are estimated from published sources and checked against them, and 5 are scenario knobs with no independent support. The range column is propagated through the parameter ensemble of Section 5 and the boundary grid of Section 6; it is a stated plausible interval across settings, not a confidence interval from a single study.

Sources for the estimated rows

mainstream persistence

Parent–child correlation in completed family size. Murphy [1999] reports values mostly 0.15–0.20 in the larger recent British and American samples reviewed; Murphy [2013], covering 46 populations in 28 countries, reports unweighted regional means from 0.09 to 0.19. The interval 0.05–0.30 is a deliberately conservative cross-setting range rather than one study's confidence interval. This is combined genetic and cultural continuity, not a heritability.

mainstream propensity cv

Computed from 43 pinned country files of the Cohort Fertility and Education database [Wittgenstein Centre, 2017]; among the 19 with mean completed fertility at or below 2.2 children, the country-median coefficient of variation is 0.570 and the observed range is 0.443–0.787. Independently cross-checked at 0.695 for United States cohorts 1947–1956 [National Center for Health Statistics, 2024]. See Section 3.1 for the tail treatment and the declared inclusion rule.

group retention/fertility/share, haredi

Regev and Gordon [2020] estimate that 13.3% of people aged 20–64 raised Haredi had left, implying 86.7% retention, with documented faction rates from 5.4% to 26.4% — which is what the wide interval reflects. Fertility 6.40 is the national statistical office's own estimate of 6.38 for 2020–2022, which Israel Democracy Institute [2024] round to 6.4 and set against a fall from 7.5 in 2003–2005; the same report gives 1.392 million Haredim in 2024, or 13.9% of Israel's population.

group retention/fertility/share, amish

Anderson and Thiehoff [2025] analyse more than 50,000 households and estimate an age-40+ retention of 84.5%, which the authors identify as the best available approximation to lifetime retention, and a total fertility rate of 6.10 from a 2002–2011 snapshot. Population share 0.115% from Young Center [2024], whose definition covers horse-and-buggy affiliations and excludes Beachy Amish and Amish Mennonites; the model inherits that boundary.

Why the knobs stay knobs

No literature check can convert a structural choice or an explicitly future path into an estimate. The number of mainstream types is a discretisation. The routing of group leavers has no direct measurement. The rate at which a group's fertility advantage decays as it stops being unusual has documented instances but no established general value. And the two environment parameters describe a future that has not happened.

The last of these is the reason Section 6 exists. Fitting the environmental decline to observed fertility would consume exactly the evidence the mechanism is meant to explain, and setting it by judgement would let it control a headline it cannot support — as Figure 2 shows it doing, at 2.79 billion against the mechanism's 1.82 billion.

B Reproducibility

What produced each number in this paper

Every numerical value in the text, tables and figures is written by a script into a machine-readable result file, and from there into a LaTeX macro or a figure. No headline number is typed into the prose. The generator refuses to build if a result file is missing, if a key it expects is absent, or if a value has moved outside a stated sanity bound, on the reasoning that a paper compiling with a silently wrong number is worse than one that does not compile.

Data

Every downloaded file is checksummed and its manifest committed, so a source that is silently reissued fails loudly rather than changing a result. The inputs are World Population Prospects 2024 [United Nations Population Division, 2024a], the archived World Population Prospects revisions used for the backtest [United Nations Population Division, 2024b], the published Bayesian fertility, mortality and migration trajectories [BayesPop Project, 2024, Azose and Raftery, 2015],

Result	Produced by `scripts/`	Stored in
Engine validation	`validate_engine.py`	`out/validation_wpp2024`
Long-run diagnostics	`run_to_2150.py`	`out/run_to_2150`
Selection grid and ensemble	`run_phase5.py`	`out/phase5`
Ladder and boundary	`analyze_selection_ break_even.py`	`reference/selection_ break_even_sensitivity`
Paired-migration boundary	`analyze_paired_ selection_boundary.py`	`reference/paired_selection_ boundary_sensitivity`
Family-size dispersion	`analyze_cfe_dispersion.py`	`reference/mainstream_ propensity_cv`
Project extension	`run_un_extension.py`	`reference/un_project_ extension_summary`
Uncertainty decomposition	`decompose_uncertainty.py`	`out/uncertainty_ decomposition`
Conventional comparator	`run_uw_ensemble.py`	`out/uw_ensemble`
Historical backtest	`run_backtest.py`	`out/backtest`
This paper's macros and tables	`build_paper_results.py`	`paper/generated/`
This paper's figures	`plot_paper_figures.py`	`paper/figures/`

Table 4: The path from a model run to a number on the page. Stored results are JSON; `reference/` abbreviates `data/reference/`, which is committed to the repository, while `out/` is regenerable and is not.

the Cohort Fertility and Education database [Wittgenstein Centre, 2017], and the national cohort fertility tables [National Center for Health Statistics, 2024].

Two data sets are not redistributed. The cohort fertility tabulations are used under terms that do not permit republication, so the repository commits the fetching script, the checksums, the derived per-country statistics and the analysis — everything needed to reproduce the number — but not the source tables. The large trajectory archives are likewise fetched rather than committed.

Conventions that produce plausible wrong answers if guessed

- Ages run 0–100 where 100 is an open-ended group; sex 0 is female.
- Survival $s[a]$ is survival *into* age a, following the source's own convention, so that no translation step stands between the published table and the engine. $s[0]$ is birth survival, and $s[100]$ applies to the sum of the 99-year-olds and the existing 100+ group.
- Populations are people. The source publishes thousands; the multiplication happens exactly once, at ingest.
- Fertility is births per woman per year. The source publishes per thousand.
- Childbearing ages are 10–54. The single-age source file stops at 49 and the five-year file does not; the missing mothers are about 0.3% of world births.
- Age-specific fertility uses mid-year women, so the denominator is the mean of the 1 January figure and the following one. Getting this wrong biases births by roughly half a year of cohort change — negligible in one year, not negligible after 126.
- World quantiles are computed from each draw's own world total, never by summing country quantiles. Adding every country's 5th percentile assumes all of them land in their own bad tail in the same draw; at 2150 that shortcut gives 3.13 billion where the correct world 5th percentile is 6.97 billion.

Stored predictions

Projections intended to be scored later are written as immutable dated records with their own checksums; the writer raises rather than overwrite an existing record, and has no override flag. The comparator of Section 7.4 is stored with every quantity marked as not a claim of this project, and was deposited before the mechanism results in Section 5 existed.

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